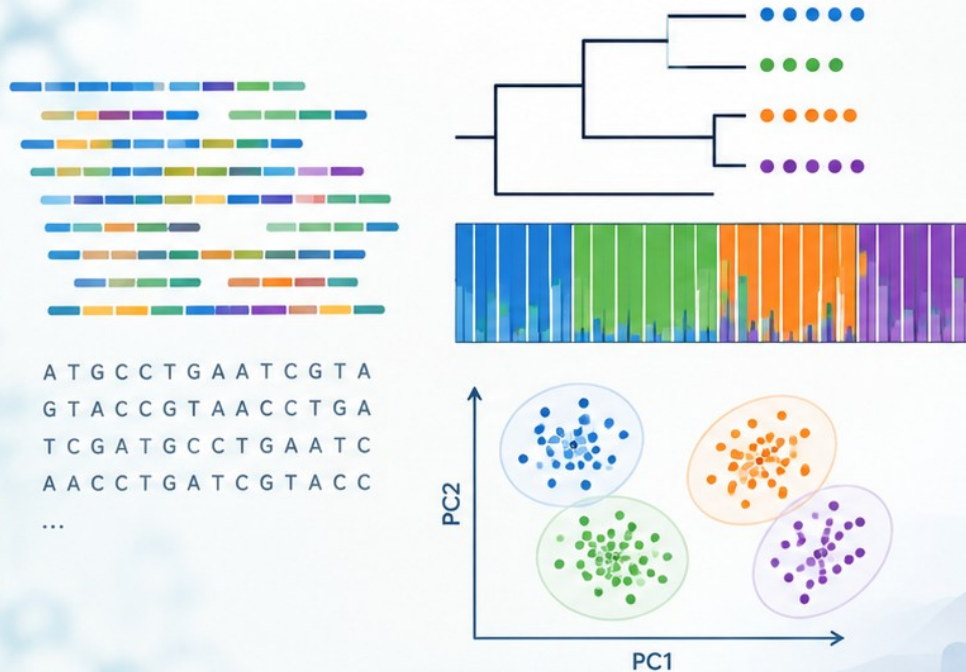


Introgression detection with D-statistic using Neanderthal genomics

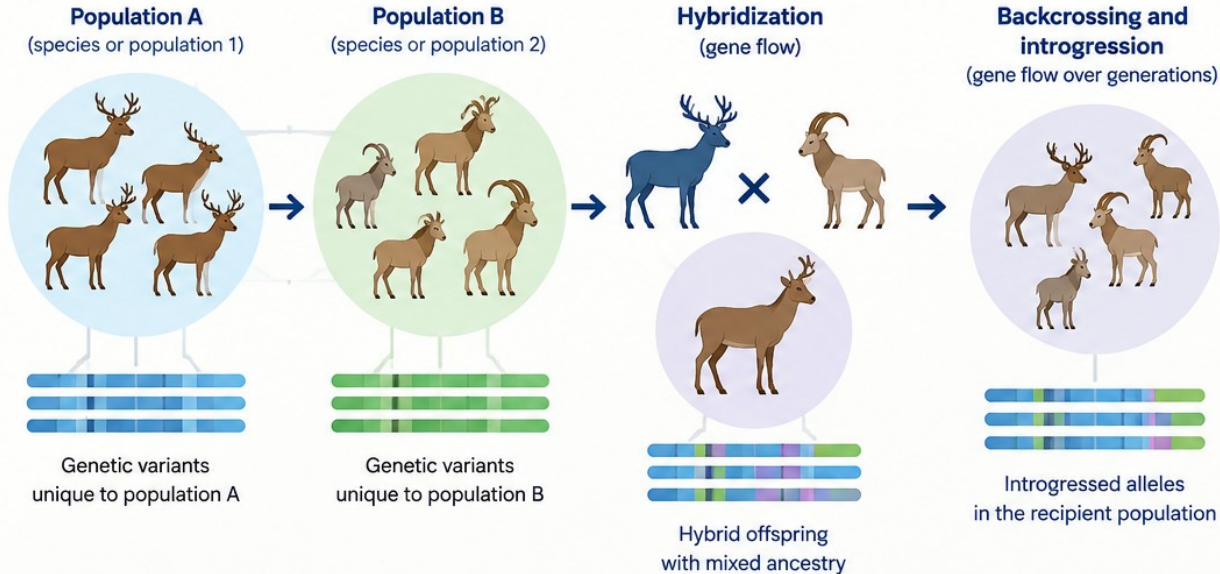
PR Stats Course

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



Genetic Introgression

The transfer of genetic material between diverged populations or species through hybridization and backcrossing.



Possible outcomes



Adaptive introgression

Introduced alleles provide a fitness advantage in the recipient population.



Neutral introgression

Alleles spread without a clear fitness effect.



Ongoing gene flow

Continued hybridization maintains mixed ancestry over time.

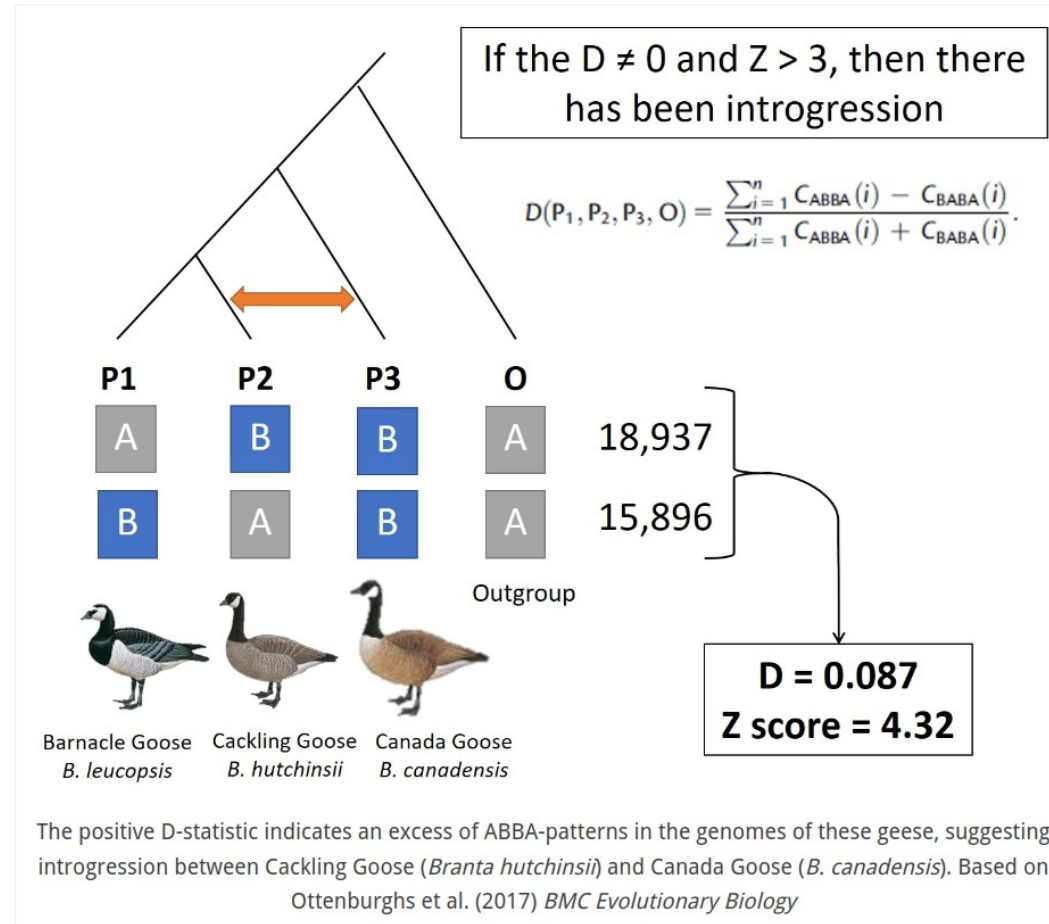


Why is introgression important?

- Introduces new genetic variation into a population.
- Can help populations adapt to changing environments.
- Can blur species boundaries and shape the evolutionary history of lineages.

Introgression shows that species are not always completely isolated — their genomes can still exchange information across generations.

The figure below illustrates the D-statistic with an example from my own work (see [Ottenburghs et al. \(2017\)](#) for more details). Comparing the genomes of four goose species reveals that Cackling Goose (*Branta hutchinsii*) and Canada Goose (*B. canadensis*) share more derived alleles than expected by chance. The resulting positive D-statistic suggests introgression between these species, which is not that surprising because there is a **hybrid zone** between these geese.





Volume 36, Issue 12
December 2019

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Supplementary Material

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JOURNAL ARTICLE

A Three-Sample Test for Introgression FREE

Matthew W Hahn , Mark S Hibbins [Author Notes](#)

Molecular Biology and Evolution, Volume 36, Issue 12, December 2019, Pages 2878–2882,

<https://doi.org/10.1093/molbev/msz178>

Published: 02 August 2019

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Abstract

Many methods exist for detecting introgression between nonsister species, but the most commonly used require either a single sequence from four or more taxa or multiple sequences from each of three taxa. Here, we present a test for introgression that uses only a single sequence from three taxa. This test, denoted D_3 , uses similar logic as the standard D -test for introgression, but by using pairwise distances instead of site patterns it is able to detect the same signal of introgression with fewer species. We use simulations to show that D_3 has statistical power almost equal to D , demonstrating its use on a data set of wild bananas (*Musa*). The new test is easy to apply and easy to interpret, and should find wide use among currently available data sets.

Keywords: [hybridization](#), [ABBA-BABA](#), [Patterson's \$D\$](#) , [gene flow](#), [admixture](#)

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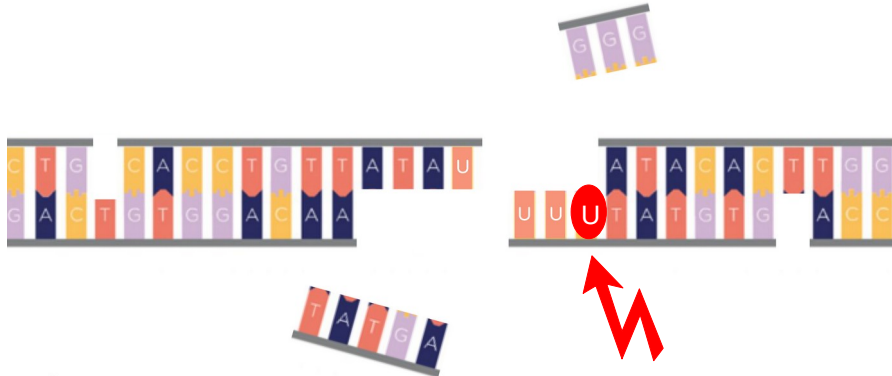
Introduction to ancient DNA data analysis ...very brief and just in case :))



TTCAGGAAAACCCGATAAAAATTCTTTATTGGGGGAGGGGCTCAAACAAGAAAATAATCAACAAGTGGTGTCCAGAGTGGAGCCAGGG
CCTCCTGGGGACAGCAAGACTGCCTGGGGAGCGGGAAGCAGCTCCCCGTCTCTGGGGGAGGCGTGGCTGGAGGGGAGGCTGGACCA
CAGGAGGGCAGCGCCCTGGGCAACCCTATGTAGATGAAGCTGCCGAGAGGATCAAAGAACCAGACAGGAGGAAAGAGGCGGTGAAG
TCTCCTGTCTCATCCCTTAGGAAGCCTGAGGAGATGGGTAAGGGCATTAGAAAGCCTCGAACCCAGGGGCAGAGGATAGTTGTAGGCA

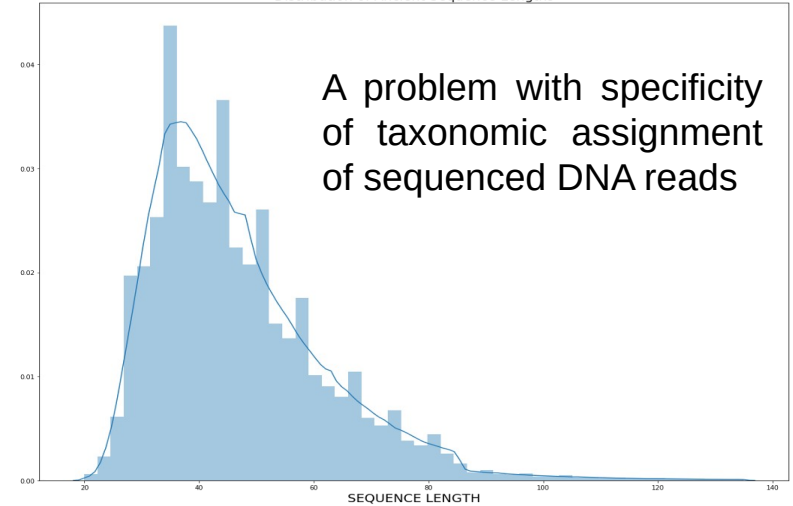
Peculiarity: very few samples with little amounts of DNA, hence hard for statistical analysis
Low statistical evidence to make confident conclusions, predictions can rarely be validated

Ancient DNA (aDNA) degradation



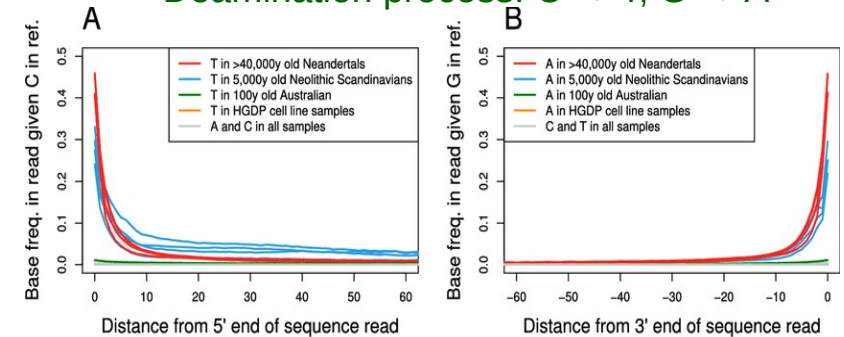
DNA fragmentation

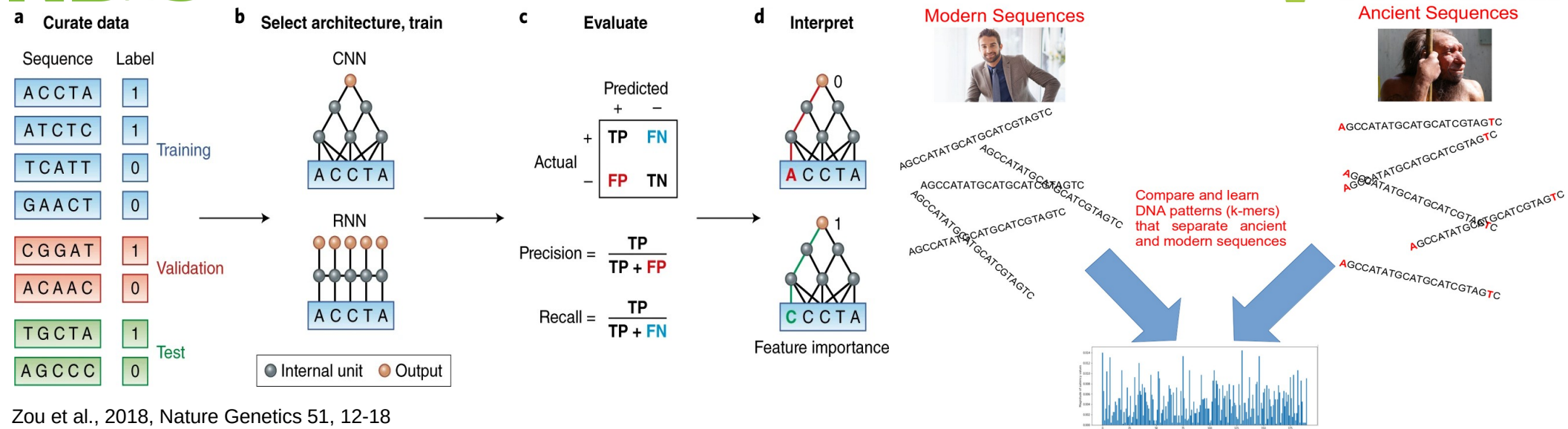
Distribution of Ancient Sequence Lengths



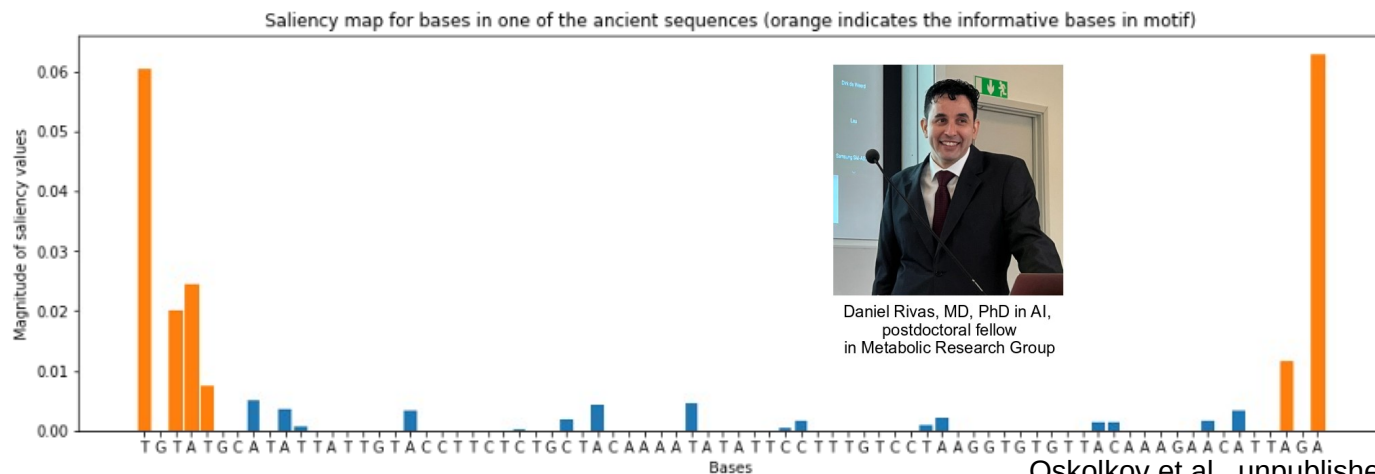
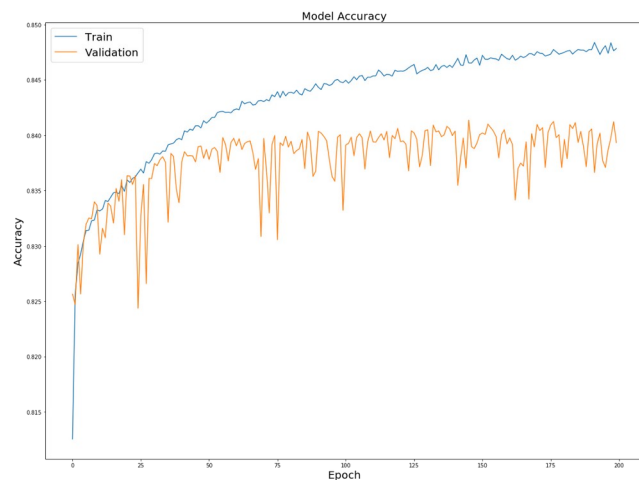
Damage / deamination pattern

Deamination process: $C \rightarrow T$, $G \rightarrow A$





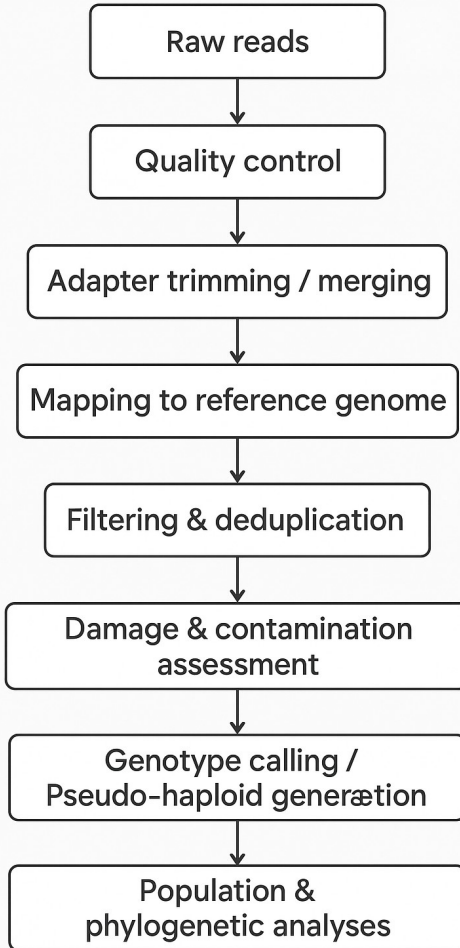
Zou et al., 2018, Nature Genetics 51, 12-18



Oskolkov et al., unpublished



Ancient DNA (aDNA) Analysis Workflow



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Primer | Published: 11 February 2021

Ancient DNA analysis

[Ludovic Orlando](#) , [Robin Allaby](#), [Pontus Skoglund](#), [Clio Der Sarkissian](#), [Philipp W. Stockhammer](#), [María C. Ávila-Arcos](#), [Qiaomei Fu](#), [Johannes Krause](#), [Eske Willerslev](#), [Anne C. Stone](#) & [Christina Warinner](#)

[Nature Reviews Methods Primers](#) 1, Article number: 14 (2021) | [Cite this article](#)

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Abstract

Although the first ancient DNA molecules were extracted more than three decades ago, the first ancient nuclear genomes could only be characterized after high-throughput sequencing was invented. Genome-scale data have now been gathered from thousands of ancient archaeological specimens, and the number of ancient biological tissues amenable to genome sequencing is growing steadily. Ancient DNA fragments are typically ultrashort molecules and carry extensive amounts of chemical damage accumulated after death. Their extraction, manipulation and authentication require specific experimental wet-laboratory and dry-laboratory procedures before patterns of genetic variation from past individuals, populations and species can be interpreted. Ancient DNA data help to address an entire array of questions in anthropology, evolutionary biology and the environmental and archaeological sciences. The data have revealed a considerably more dynamic past than previously appreciated and have revolutionized our understanding of many major prehistoric and historic events. This Primer provides an overview of concepts and state-of-the-art methods underlying ancient DNA analysis and illustrates the diversity of resulting

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[AUTHORS INFO & AFFILIATIONS](#)

<https://doi.org/10.1093/mSystems/03035-21> | [Check for updates](#)

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ABSTRACT

Like modern metagenomics, ancient metagenomics is a highly data-rich discipline, with the added challenge that the DNA of interest is degraded and, depending on the sample type, in low abundance. This requires the application of specialized measures during molecular experiments and computational analyses. Furthermore, researchers often work with finite sample sizes, which impedes optimal experimental design and control of confounding factors, and with ethically sensitive samples necessitating the consideration of additional guidelines. In September 2020, early career researchers in the field of ancient metagenomics met (Standards, Precautions & Advances in Ancient Metagenomics 2 [SPAAM2] community meeting) to discuss the state of the field and how to

ANNUAL REVIEWS

Annual Review of Genomics and Human Genetics

A Robust Framework for Microbial Archaeology

Christina Warinner,^{1,2} Alexander Herbig,¹ Allison Mann,^{1,2} James A. Fellows Yates,¹ Clemens L. Weiß,³ Hernán A. Burbano,³ Ludovic Orlando,^{4,5} and Johannes Krause¹

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²Department of Anthropology, University of Oklahoma, Norman, Oklahoma 73019

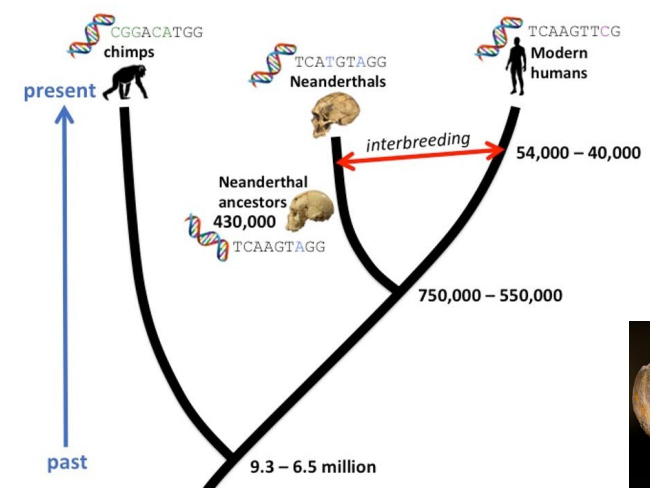
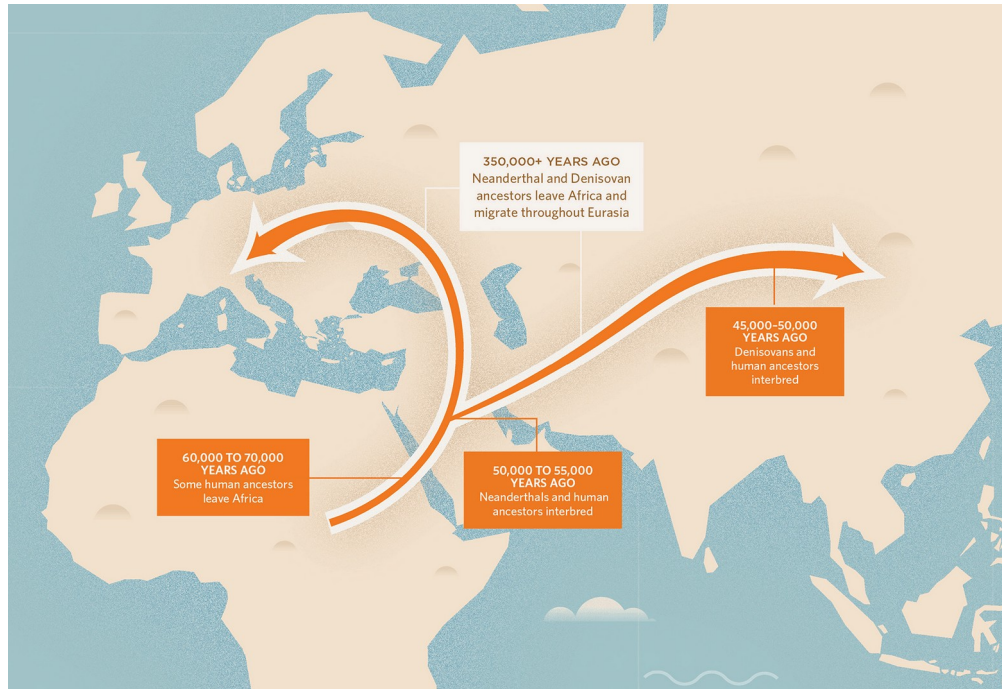
³Research Group for Ancient Genomics and Evolution, Department of Molecular Biology, Max Planck Institute for Developmental Biology, Tübingen 72076, Germany

⁴Centre for GeoGenetics, Natural History Museum of Denmark, 1350 Copenhagen K, Denmark

⁵Laboratoire d'Anthropologie Moléculaire et d'Imagerie de Synthèse, CNRS UMR 5288, Université Toulouse III – Paul Sabatier, Toulouse 31000, France

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Neanderthal introgression in modern human populations

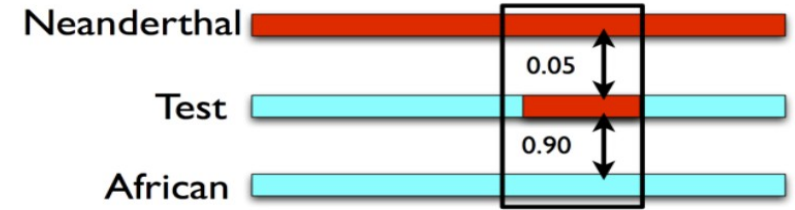
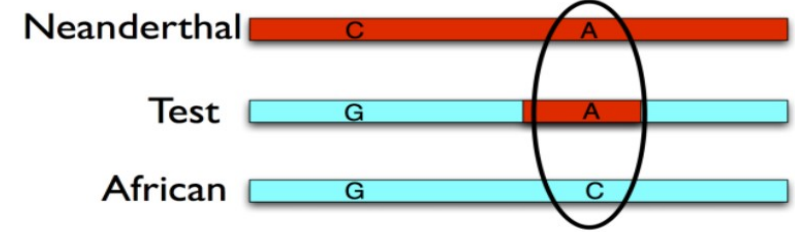
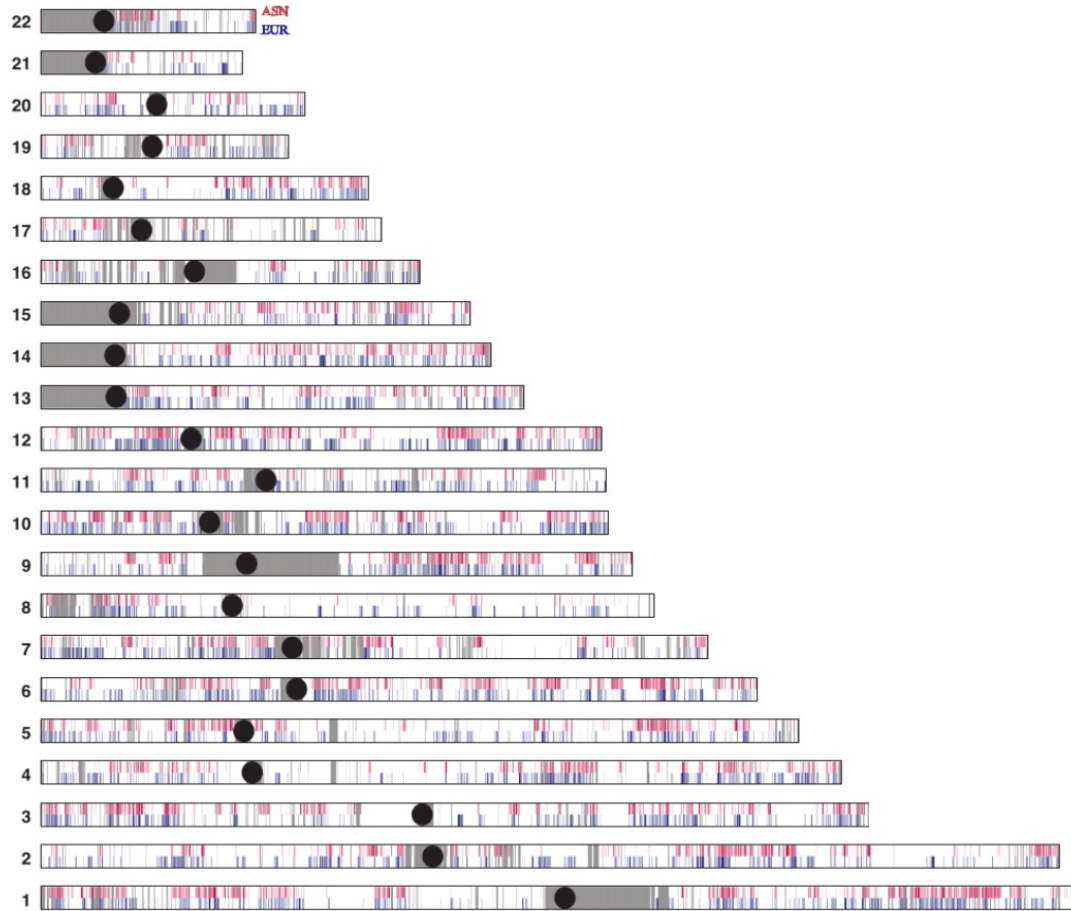


Neanderthal DNA in Modern Human Genomes Is Not Silent

From skin color to immunity,
human biology is linked to our archaic ancestry

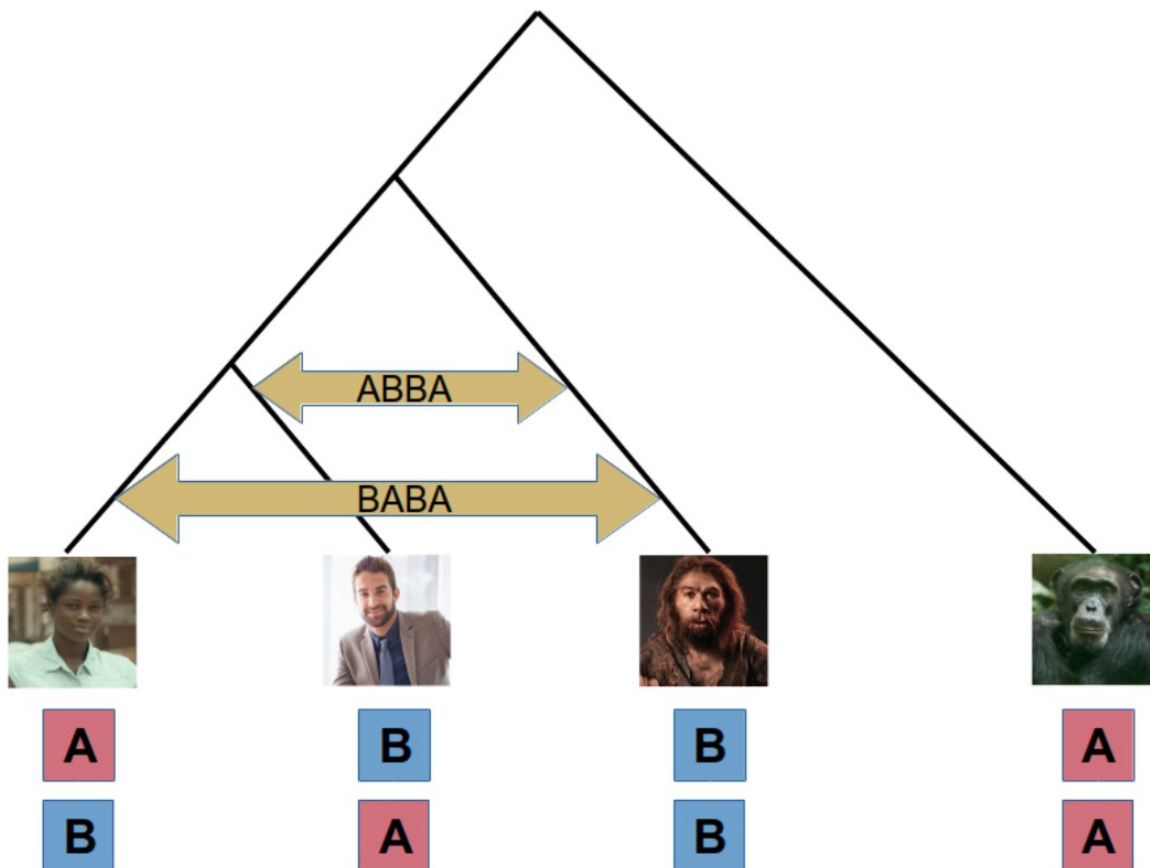


2010 Draft Neanderthal Genome Sequenced



Conditional Random Fields (CRF): David Reich et al., Nature 2014, 2016
 S* and S' statistics: Benjamin Vernot and Joshua Akey, Science 2014, 2016

F4-statistic, D-statistic: ABBA-BABA test



- Four populations considered: Ape is an outgroup, Europeans and Africans are tested for their gene flow with Neanderthals
- A – ancestral allele, B – derived allele
- Count number of sites corresponding to ABBA and BABA situations

$$D = \frac{N_{ABBA} - N_{BABA}}{N_{ABBA} + N_{BABA}}$$

- N_{ABBA} the total counts of ABBA patterns, N_{BABA} the total counts of BABA patterns
- Excess of shared derived alleles between Europeans and Neanderthals indicates a gene flow between them
- Positive values of D imply Neanderthals are closer to Europeans
- Negative values of D imply Neanderthals are closer to Africans

Example of D-statistic computation from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute D-statistic in order to test potential Neanderthal introgression to European (CEU) and African (YRI) populations from the 1000G project.

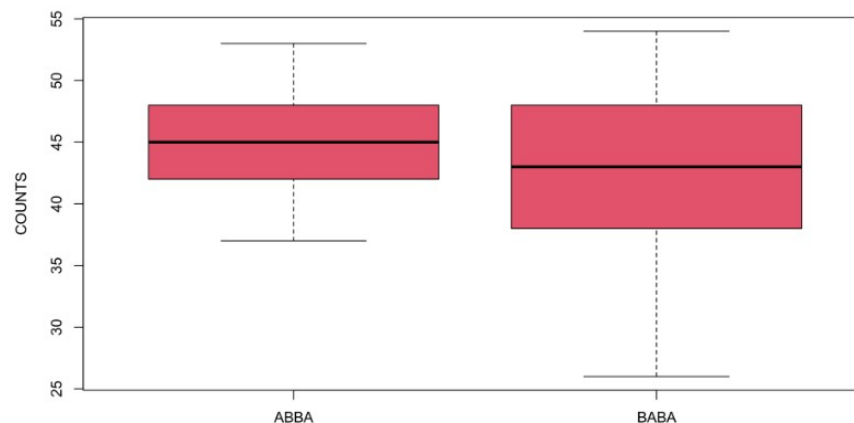
```
1 echo Neandertal.bam > bams
2 find $PWD -name '*YRI*.bam' | head -10 >> bams
3 find $PWD -name '*CEU*.bam' | head -10 >> bams
4 angsd -out out -doAbbababa 1 -bam bams -doCounts 1 -anc anc.fa
5 Rscript jackKnife.R file=out.abbababa indNames=bams outfile=out_abbababa_results
```

and, finally, we will display the resulting table of D-statistics and plot the ABBA and BABA counts using custom R scripts:

```
1 df <- read.delim(file.path(path, "out_abbababa_results.txt"), header = T, sep = "\t")
2 df <- df[grepl("Neandertal", as.character(df$H3)), ]
3 df <- df[grepl("CEU", as.character(df$H2)), ]
4 df <- df[grepl("YRI", as.character(df$H1)), ]
5 YRI <- matrix(unlist(strsplit(as.character(df$H1), "\\.")), ncol = 9, byrow = T)
6 df$H1 <- paste(YRI[, 6], YRI[, 2], sep = "_")
7 CEU <- matrix(unlist(strsplit(as.character(df$H2), "\\.")), ncol = 9, byrow = T)
8 df$H2 <- paste(CEU[, 6], CEU[, 2], sep = "_")
9 df$H3 <- "NEAND"
10 df[1:8, 1:8]
```

	H1	H2	H3	nABBA	nBABA	Dstat	jackEst	SE
11	YRI_NA19102	CEU_NA12342	NEAND	51	46	0.05154639	0.05154639	0.1220787
12	YRI_NA19213	CEU_NA12342	NEAND	38	31	0.10144930	0.10144930	0.1286760
13	YRI_NA18504	CEU_NA12342	NEAND	50	53	-0.02912621	-0.02912621	0.1353893
14	YRI_NA19108	CEU_NA12342	NEAND	38	42	-0.05000000	-0.05000000	0.1393287
15	YRI_NA19238	CEU_NA12342	NEAND	44	38	0.07317073	0.07317073	0.1305890
16	YRI_NA19102	CEU_NA12348	NEAND	46	45	0.01098901	0.01098901	0.1076116
17	YRI_NA19213	CEU_NA12348	NEAND	37	34	0.04225352	0.04225352	0.1122396
18	YRI_NA18504	CEU_NA12348	NEAND	42	49	-0.07692308	-0.07692308	0.1068372

```
1 boxplot(df$nABBA, df$nBABA, names = c("ABBA", "BABA"), ylab = "COUNTS", col = 2)
```



```
1 wilcox.test(df$nABBA, df$nBABA)
```

Wilcoxon rank sum test with continuity correction

data: df\$nABBA and df\$nBABA

W = 367, p-value = 0.2933

alternative hypothesis: true location shift is not equal to 0

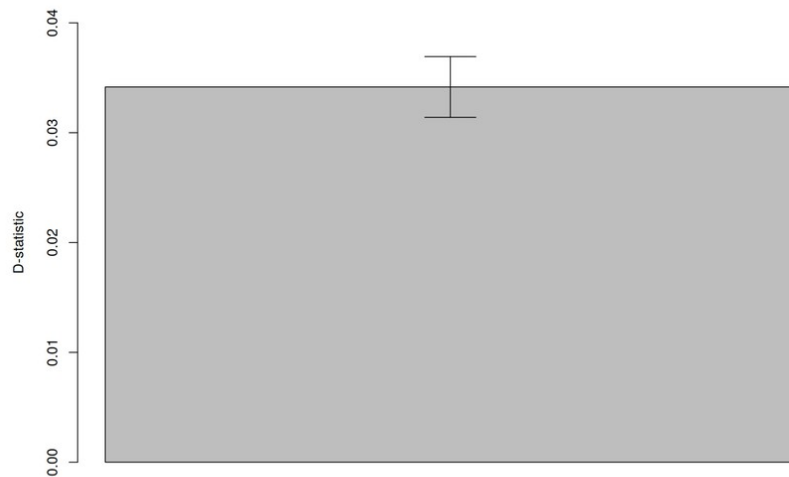
D-statistic computation for higher coverage samples

D-statistic analysis requires large number of genomic sites, therefore the low-coverage 1000G data are not optimal. Now, we are going to use French and Yoruba samples from the **Draft Neanderthal genome** project where Neandertal introgression was first detected:

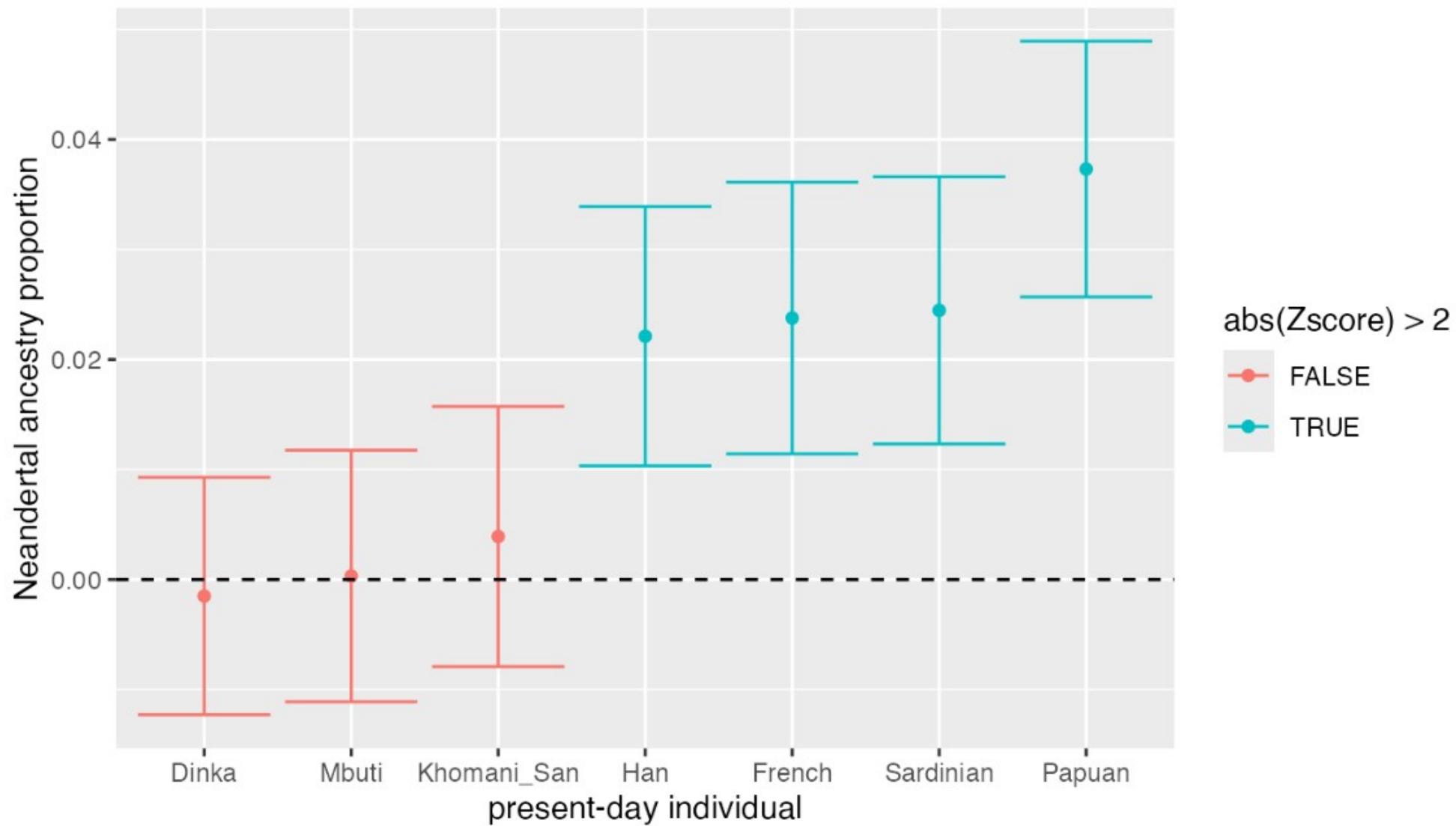
	H1	H2	H3	nABBA	nBABA	Dstat	jackEst
1	Yoruba.bam	French.bam	Neandertal.bam	267387	249721	0.03416308	0.03416308
2	Neandertal.bam	French.bam	Yoruba.bam	732198	249721	0.49136130	0.49136130
3	Neandertal.bam	Yoruba.bam	French.bam	732198	267387	0.46500400	0.46500400

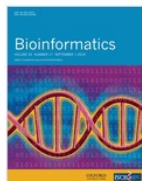
	SE	Z
1	0.002764548	12.35756
2	0.002449853	200.56760
3	0.002868187	162.12470

First line of the output corresponds to the tree displayed on the previous slide, i.e. where we test for potential gene flow from Neanderthals to modern Europeans or Africans. Large number of genomic sites ensures confident positivity of the D-statistic.



Positive D-statistic with errors bars not overlapping zero implies a gene flow between Neanderthals and modern Europeans.





Volume 35, Issue 17
September 2019

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3 Example usage

4 Additional functionality

Acknowledgements

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Supplementary data

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JOURNAL ARTICLE

admixr—R package for reproducible analyses using ADMIXTOOLS

Martin Petr, Benjamin Vernot, Janet Kelso

Bioinformatics, Volume 35, Issue 17, September 2019, Pages 3194–3195, <https://doi.org/10.1093/bioinformatics/btz030>

Published: 22 January 2019 Article history

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Abstract

Summary

We present a new R package *admixr*, which provides a convenient interface for performing reproducible population genetic analyses (f_3 , D , f_4 , f_4 -ratio, *qpWave* and *qpAdm*), as implemented by command-line programs in the ADMIXTOOLS software suite. In a traditional ADMIXTOOLS workflow, the user must first generate a set of text configuration files tailored to each individual analysis, often using a combination of shell scripting and manual text editing. The non-tabular output files then need to be parsed to extract values of interest prior to further analyses. Our package simplifies this process by automating all low-level configuration and parsing steps, making analyses as simple as running a single R command. Furthermore, we provide a set of R functions for processing, filtering and manipulating datasets in the EIGENSTRAT format. By unifying all steps of the workflow under a single R framework, this package enables the automation of analytic pipelines, significantly improving the reproducibility of population genetic studies.



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